

Comments on “Z-residuals for Diagnosing Bayesian Models”

This is an interesting piece of work. The authors propose a Z-residual concept that applies to general Bayesian models. The following issues need to be addressed. The major issues are #3, #4, #6, #7.

1. Page 3, line 36. Can the authors explain why you use the term “Z-residual”?
2. Page 3, line 40. Can the authors clarify if U_i is needed in the randomized survival probability when the outcome variable Y is continuous?
3. (How to position this paper?) If U_i is not needed when Y is continuous, the authors need to clarify the main contribution of the paper. When Y is continuous, is your residual still novel in the Bayesian literature? Does the significance still remains? If not, the authors need to reposition this paper by saying that the novelty resides in the discrete case.
4. (Lack of sensitivity analysis) Does the choice of the prior influence diagnosis? In particular, when the sample size is small, an ill-chosen prior may lead to a wrong rejection of the model. Am I right? The authors should perform sensitivity analysis.
5. Page 17, Figure 1. I am not sure about the value of plotting Z-residuals against the index. What insights can the plots offer in addition to those offered by the QQ plots?
6. (Lack of numerical studies wrt other Bayesian models) The authors claim that they proposed “general-purpose residuals”. But the numerical studies only focus on (hurdle) negative binomial models. This is relevant to my comment #3. If the authors position this paper as a general methodology, then more numerical studies using different models (for continuous and discrete data) should be presented.
7. (Double use of the data) I like the point the authors make about the double use of data in fitting and diagnosing the model. My question is: although you do not use y_i , you do use the rest of the data. Why this does not count as “double use of the data”. Furthermore, y_i is generated from the same model that generates other y 's. So I believe your approach still has the same problem. The importance-sampling approach presented in Section 2.4 may alleviate the severity of the problem.
8. (Insufficient simulation replications) Page 20, Figure 3. The authors should increase the simulation replications to 1000 if type I errors are part of the results.
9. (Inflated type I error) Page 20, Figure 3(a). Please report type I errors in the numerical form. By reading from the graphical presentation, I feel that some of type I errors are inflated, close to 0.1. The same can be said to Figure 6.

10. (Bin size in Figure 4) How do the authors determine the bin size? The analysis of variance (and its p-value) is heavily dependent on how you bin the residual points.
11. (Tone down the self-promoting statements) Page 15, line 40: “very promising”. Page 30, line 26, “excellent performance”. Please use objective statements.
12. (Reduce the excessive use of uncommon abbreviations) The authors used GOF, PPC, PPP, RSP, LOO, ISCV, NB, HNB, MSP, SW, PIT, AOV, and other uncommon abbreviations. They do make the reading difficult. I have to flip the pages and spend a lot of time to find what they really mean. The journal Biometrika explicitly forbid the use of uncommon abbreviations. Please remove as many as possible.